Analysis Choices and Ranking Stability in Kinase Inhibitor Selectivity Scores

Polina Vinogradova*,†,‡

†*Independent Researcher*

‡*ORCID: 0000-0003-3271-3841*

E-mail: polina.vino@gmail.com

Abstract

Kinase selectivity scores can rank the same compounds differently. We compare four common scores across four profiling datasets. Agreement changes with affinity weighting, the compounds ranked and the treatment of pairs without detected binding. We relate ranking sensitivity to binding profile features. Moving the activity boundary reorders compounds under every score, and the boundary must be fixed by protocol. We propose four criteria for selectivity scores and, with Shannon's grouping rule, derive from them a gated entropy with exponential affinity weights. We prove the range of weighting for which adding a weakly bound target cannot make a compound appear more selective. Scores that follow the share of targets with detected binding recover rankings from fewer targets than the ratio and the published entropy, but on small panels many compounds with few active targets cannot be scored. These findings support reporting analysis choices, ranking sensitivity and compound coverage.

Keywords: binding selectivity, cancer drug discovery, cheminformatics, kinase inhibition, research methodology.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with 94 FDA-approved agents and several hundred in clinical development [1]. The human kinome comprises more than 500 proteins with a conserved ATP-binding site [2], and selectivity is a primary criterion in lead optimization [3]. Profiling assays measure a compound against a panel of tens to hundreds of kinases, and the resulting profile is often reduced to one selectivity score. Several scores are in use:

- the S-score [4], the fraction of the panel bound below an affinity cutoff
- the selectivity entropy [5], the Shannon entropy of the profile treated as a distribution
- the Gini coefficient [6] of the profile
- the partition index [7], the share of binding that falls on one target
- ratio-based measures, which compare the primary target with an off-target.

They are often used interchangeably, although each summarizes a different aspect of the profile.

Earlier work established that these scores can disagree. Bosc et al. [8] compared six scores on the Davis and Anastassiadis datasets, computed several at more than one parameter value, compared them on compounds and kinases shared between datasets, and found that compound rankings differ substantially between them. They attributed individual disagreements to profile shape, such as a broad binder with one dominant target. The KInhibition platform [9] noted that conflicting results across metrics are common, and introduced a composite score. A later resource [10] from the same group offers a choice of scores. Miljkovic and Bajorath [11] found the concentration-dependent selectivity score of Klaeger et al. [12] broadly consistent with activity data from ChEMBL. Karaman et al. [4] and Uitdehaag et al. [13] showed that the S-score and the entropy of individual compounds vary when they are computed on random subpanels.

What has not been characterized systematically is how analysis choices alter agreement between the scores. These choices include how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which assay targets count as kinases, and where each score's parameter is set. It is also not known which associations between profile shape and ranking instability persist across these choices, or how much of a ranking a smaller panel retains once some compounds can no longer be scored.

This work adds a systematic account of these choices. We compare the S-score, entropy, Gini and ratio on four datasets that span three assay technologies. Each is curated to one reference target per human protein kinase, and the entropy and Gini are computed in both a common log-affinity form and their published forms. We vary the compounds ranked, the treatment of pairs without detected binding and each score's parameter, and relate the resulting instability to features of the binding profile. From these results we state four criteria for a selectivity measure: activity eligibility, locality of the activity boundary, monotonicity when a weakly bound kinase is added, and orientation on profiles with equally strong targets. We evaluate the existing scores and simple comparators against them. We then derive a gated entropy that satisfies all four. Among scores that weight each target by its distance above the activity boundary and combine the weights with Shannon's grouping rule, the criteria leave one family with a single parameter, the *scale*. The scale is the readout difference over which a weight grows by a factor of e. Monotonicity fixes the range of the scale. Within that range, the scale can be chosen so that the score recovers rankings from small panels as fast as the share of targets with detected binding. Stating such criteria follows the axiomatic approach Shannon[14] used for entropy. We measure how well each score's ranking is recovered from random subpanels, and for how many compounds, then give recommendations for reporting selectivity scores.

Methods

Datasets, Target Selection and Missing Values

Table 1 summarizes the four datasets. The Davis dataset[15] reports dissociation constants from a competition-binding assay, and the Klaeger dataset[12] apparent dissociation constants from Kinobeads chemoproteomics in cell lysate (beads carrying kinase inhibitors that capture kinases from the lysate, with a mass-spectrometry readout). Both were converted, as in DeepDTA[16], to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$. The Anastassiadis dataset[17] reports residual kinase activity in a radiometric assay at 0.5 μM inhibitor, which we converted to percent inhibition (100 minus residual activity, clipped to $[0, 100]$). The Metz dataset[18] reports $\mathrm{p}K_i$ values from a medicinal-chemistry campaign. Because the assays and readouts differ, each score is computed on its dataset's own readout, and datasets are compared only through the rankings computed within each.

Table 1: The four datasets. The floor x_0 is the value given to pairs with no detected binding (Davis, Klaeger) or not tested (Metz). On Anastassiadis it is an operational cutoff for weak inhibition. τ^* is the activity threshold.

Dataset	Assay; readout	Compounds	Targets (released, curated)	x_0	τ^*
Davis	competition binding; $\mathrm{p}K_d$	68	433, 363	5.0	6
Klaeger	Kinobeads in lysate; $\mathrm{p}K_d$	222	343, 217	5.0	6
Anastassiadis	radiometric activity; % inhibition	178	300, 291	20%	50%
Metz	biochemical activity; $\mathrm{p}K_i$	704	172, 172	4.0	6

Target selection. Selectivity across kinases is expressed as a single value, so each kinase should be counted once. A panel that lists several constructs (recombinant forms of a protein, such as mutants or single domains) of one gene lists that kinase several times. The released Davis panel lists 433 constructs. They include mutant, phosphorylation-state and single-domain constructs of the same gene (12 for EGFR), 20 lipid kinases and three non-human kinases, and for 26 of the 68 compounds the two best-bound constructs are forms of one gene. We keep one reference construct per human protein-kinase gene: the wild-type construct, or

its catalytic or N-terminal kinase domain where several domains are profiled. Genes profiled only as mutants are removed. The 343 Klaeger targets are all proteins that the beads captured and mass spectrometry measured, including complex subunits, such as cyclins, that bind the captured kinases. We keep the single-gene protein kinases that Klaeger et al. annotate, except Q6ZSR9 (which UniProt lists as an uncharacterized protein without a kinase domain), which leaves 217. This removes lipid kinases, four protein groups, and BCR and TRRAP, which the source does not annotate as kinases and flags as possibly co-enriched with ABL1 and CDK8. On Anastassiadis we keep the first-listed of nine kinases profiled in two complexes or isoforms. The Metz columns are already one human protein kinase each. We call the targets kept on each dataset its *curated panel* (Table 1). The code repository lists every target with the reason it was kept or removed, and Supporting Information (Section S1) compares the released panels.

Floor-coded and missing values. On Davis, Klaeger and Metz, several kinds of pairs receive the same floor value x_0, which is not a measured detection limit. On Davis, the released matrix assigns $\mathrm{p}K_d = 5.0$ ($K_d = 10$ μM) to pairs with no detected binding, following DeepDTA[16]. Bosc et al.[8] used nearly the same value (10.1 μM). On Klaeger, we kept the 5,123 drug–target pairs that the original authors classified as high confidence. We coded at $\mathrm{p}K_d = 5.0$ three kinds of pairs: those with no detected binding, those whose fit was classified as low confidence, and the 194 measured affinities weaker than 10 μM. A low-confidence fit does not establish weak binding: of the 405 such pairs on curated targets, 375 have a fitted $\mathrm{p}K_d$ above 5. Giving all of these pairs the floor value is an analysis convention. On Metz, pairs that were not tested are coded at $\mathrm{p}K_i = 4.0$, the lowest value reported in the data. A median Metz compound is measured against only 21 of the 172 kinases, so we kept the 704 compounds measured against at least 50 (median 64). On Anastassiadis, the 1.1% of missing entries were set to 0% inhibition. Most pairs on this panel show little inhibition: 85% of the entries are at or below 20%, and each 5% interval above 20% holds less than 3% of the

entries. We place x_0 at 20%, the upper end of these low values, as an operational cutoff for weak inhibition.

A compound is *active* against a target above the activity threshold τ^*: pK_d or pK_i 6 (K_d or K_i of 1 μM) on the affinity datasets, and 50% inhibition on Anastassiadis. With a Hill coefficient of 1, 50% inhibition at 0.5 μM inhibitor means an IC_{50} of 0.5 μM, close to the 1 μM of the other datasets. Supporting Information (Section S1) repeats the Davis comparison of the scores with the threshold one log unit higher. An *active compound* has at least one target above τ^*.

Selectivity Scores and Their Implementations

A compound's profile is the vector $\mathbf{x} = (x_1, \ldots, x_N)$ of its readouts across the N panel targets, with higher values for tighter binding or stronger inhibition. Every score is oriented so that higher values are more selective (Table 2).

S-score. The S-score of Karaman et al.[4] is the fraction of the panel bound above a threshold,

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[x_i > \tau], \tag{1}$$

negated. They reported K_d cutoffs of 3 μM and 100 nM, which lie within our sweep of τ.

Selectivity entropy. The selectivity entropy of Uitdehaag and Zaman[5] is the Shannon entropy $H = -\sum_i p_i \log_2 p_i$ (in bits) of the profile normalized to a distribution, negated. In its published form, $p_i = K_{a,i} / \sum_j K_{a,j}$, with association constants $K_{a,i} = 1/K_{d,i}$ ($1/K_{i,i}$ on Metz). It needs affinity constants, so it is not computed on Anastassiadis. To compute all four definitions from the same readout, we also use a log-affinity form (on Anastassiadis, percent inhibition), which weights each target by its readout above a baseline β:

$$p_i = \frac{\max(x_i - \beta,\, 0)}{\sum_j \max(x_j - \beta,\, 0)}. \tag{2}$$

The baseline is a parameter we introduce and has no published reference values.

Gini coefficient. Graczyk's Gini coefficient[6] measures how unequally binding is spread over the panel. Its published form uses percent-inhibition values. We compute it on the same baseline-subtracted weights,

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot w_{(i)}}{N\sum_{i=1}^{N} w_{(i)}} - \frac{N+1}{N}, \qquad w_{(i)} = \max(x_i - \beta,\, 0) \text{ sorted ascending}, \tag{3}$$

and on Anastassiadis also in its native form.

Unless stated otherwise, both the entropy and Gini are in their log-affinity forms.

Ratio. The ratio compares the primary target with the k-th ranked off-target,

$$R_b(k) = x_{(1)} - \max(x_{(k+1)}, b), \tag{4}$$

where $x_{(j)}$ is the j-th largest value. The level b is the lowest coded value (5.0 on Davis and Klaeger, 4.0 on Metz and 0% on Anastassiadis), except in the recovery analysis, where it is x_0, and in the boundary sweep, where it is swept. At this level $R_b(k)$ equals the ranking score of Bosc et al.[8] at rank $k+1$. We cap k at 5 because half of the Klaeger compounds have fewer than five active kinases.

Partition index. The partition index of Cheng et al.[7] is the fraction of a compound's molecules that would bind a reference target in a mixture of all panel kinases, $K_{a,\mathrm{ref}} / \sum_j K_{a,j}$. Only Klaeger names an intended target for each compound, so we use the most strongly bound target as the reference ($P_{\max}$), as Uitdehaag and Zaman[5] did. Like the published entropy, it is not computed on Anastassiadis. The CATDS score of Klaeger et al. is the same kind of fraction, for a designated target at one concentration. It needs Kinobeads dose–response curves, which the other three datasets lack, so it is not compared across datasets (Supporting Information, Section S6, compares a proxy).

Table 2: Score implementations. Parameter ranges are for the pK_d datasets. On Anastassiadis τ runs over $[50, 90]\%$ and β over $[0, 30]\%$, and on Metz over $[5.0, 7.5]$ and $[4.0, 5.5]$. The last three rows are comparators.

Score	Weights or formula	Parameter swept	No target above the parameter
S-score	fraction above τ	$\tau \in [5.5, 8.0]$	score 0
Entropy, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (top rank)
Gini, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (bottom rank)
Ratio $R_b(k)$	$x_{(1)} - \max(x_{(k+1)}, b)$	$k = 1, \ldots, 5$	uses b
Entropy, published	$K_{a,i}$	none	—
Partition index $P_{\max}$	$\max_i K_{a,i} / \sum_j K_{a,j}$	none	—
Gated entropy	gated exponential (below)	none	not scored (D1)
Active count	number above τ^*	none	not scored (D1)
Entropy, $\beta = \tau^*$	$\max(x_i - \tau^*, 0)$	none	not scored (D1)
S-score, $\tau = x_0$	fraction above x_0	none	not scored (D1)

Conventions. Where the entropy and Gini are undefined, we set both to 0. This ranks the compound as maximally selective under the entropy and minimally selective under the Gini. $R_b(k)$ uses b for a missing off-target and so scores every compound, as Bosc et al.'s ranking score does. Log-affinity weights with $\beta = x_0$ give floor-coded pairs zero weight. Association constants give them a small positive weight if they are kept at the floor value. For the published entropy and $P_{\max}$, we give them zero weight. As a sensitivity analysis for the published entropy, we keep floor-coded pairs at the floor value. On Klaeger, we also give every measured high-confidence affinity its measured value, including the 194 weaker than the floor, and the other floor-coded pairs zero weight. No data show whether floor-coded pairs should have zero weight or the floor value.

Comparators. The comparators are simple variants of existing scores. The only new score in this study is the gated entropy. The active count (the number of targets above τ^*, which is the numerator of the S-score at τ^*, with fewer more selective), the entropy with $\beta = \tau^*$ and the S-score with $\tau = x_0$ (the share of the panel with detected binding) are scored only for active compounds. Otherwise, a compound with no active target would have an active count of 0, fewer than any active compound, and would rank as the most selective.

Ranking Comparisons and Sensitivity Analysis

For each score and parameter value, compounds are ranked from most selective (rank 1), with ties given their average rank. Ranks are always computed among a stated set of compounds. Analyses of compounds with an active kinase rank them among themselves. Analyses that compare compounds with and without an active kinase rank all compounds. Agreement between two definitions is taken to be the Spearman correlation between their median rank vectors over each definition's parameter sweep: τ in steps of 0.25 (11 values) and β in steps of 0.25 (7 values) on the pK datasets, τ in steps of 5% (9 values) and β in steps of 5% (7 values) on Anastassiadis, and $k = 1, \ldots, 5$. Using mean ranks instead changes the correlations among all compounds by at most 0.08. Bootstrap 95% intervals use 1,000 resamples of compounds.

Each parameter value gives one ranking of the compounds. A compound's *rank instability* is the standard deviation of its ranks across one definition's sweep. Its *aggregated rank instability* covers all four definitions. Aggregation is done by taking the standard deviation of the compound's ranks over the rankings from all four sweeps together. Each definition therefore counts in proportion to its number of parameter values, and the S-score, with the most values, carries the most weight. We compute the Spearman correlation between within-definition instability and each profile feature: the number of active kinases, the top_1–top_2 affinity gap, the second-ranked affinity, and the standard deviation, range and skewness of the active affinities. A feature undefined for a compound with too few active kinases is set to 0. Supporting Information (Section S3) repeats the analysis with these compounds excluded. These associations are exploratory. A Bonferroni correction, checked against the Holm and Benjamini–Hochberg procedures, is applied only to the 24 tests per dataset in Supporting Information Table S4 (six features and four definitions), while other p-values are uncorrected.

To compare how much rankings change between datasets with how much they change between definitions, we matched 46 compounds profiled in both Davis and Klaeger by PubChem CID and restricted both panels to the 201 kinases they share (matched by gene symbol, with renamed symbols mapped to current ones). We then computed the Spearman

correlation between each definition's median ranking on Davis and on Klaeger, and between each pair of definitions' median rankings within each dataset.

Evaluation Criteria

A selectivity measure f is a function that, for a given profile $\mathbf{x}$ of any length N, returns a real number $f(\mathbf{x})$, with higher values for more selective profiles. Let $B = [x_0, \tau^*]$ be the interval between the floor and the activity threshold. We present four criteria. D1 decides which compounds are scored, D2 and D3 concern how their scores change when the activity boundary moves and when the panel grows, and D4 concerns the direction of the score. All four are judged by construction.

D1 (activity eligibility). $f(\mathbf{x})$ is undefined, or flagged as not eligible, whenever $\max_i x_i \leq \tau^*$. A compound with no target above τ^* has no target against which to judge selectivity. D1 is a reporting rule: a score satisfies it if its definition withholds or flags these compounds. Bosc et al.[8] withheld ranks for such compounds under the S-score only, whereas D1 applies the rule to every score.

D2 (boundary locality). Let θ be the parameter that places a score's activity boundary, i.e. the S-score's τ, the entropy's and Gini's β and the ratio's level b. If no value of $\mathbf{x}$ lies between θ and θ', then $f(\mathbf{x};\theta) = f(\mathbf{x};\theta')$. Moving the boundary should change a compound's score only when it changes which of the compound's targets lie above the boundary.

D2 allows rankings to change when the boundary passes a compound's values. To measure how much they change, we move θ across B. A pair of compounds *reverses in* B if $f(\mathbf{x};\theta) - f(\mathbf{y};\theta)$ is positive at some $\theta \in B$ and negative at another, and we report the share of the pairs of compounds scored under D1 that reverse. To count reversals, each score is evaluated at a set of values of θ in B. The S-score changes only where τ passes an observed value, so it is evaluated at every observed value in B and its percentages are exact. The other parameters are evaluated at 201 equally spaced values. Apart from the S-score's, the percentages are therefore estimates.

Some scores fix the boundary at one value, e.g. the active count, which counts the targets above τ^*. Such a score is assessed, in D2 and in the reversal counts, by moving that value over B. The value of θ is set in the analysis protocol according to the intended use, e.g. by whether binding weaker than τ^* matters at the concentration at which the compound will be used in practice. At different values of θ, a score can measure different things: on Davis, the S-score at $\tau = 5$ counts the targets with K_d below 10 μM, and at $\tau = 6$ those below 1 μM. Its rankings can reasonably differ between the two.

D3 (monotonicity under weak-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_0 < x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$. Otherwise a compound can be made to look more selective by profiling it against more kinases it binds weakly. A score satisfies D3 if adding such a target can be proved never to raise the score, and violates it if a profile can be constructed on which it does. Tests on measured profiles (Results) are secondary checks of D3.

D4 (orientation). Fix the panel size $N \geq 2$ and all of the score's parameters. For every value a above τ^* and above every cutoff and baseline the score uses, let $\mathbf{v}_m$ be the profile with m targets at a and the other $N - m$ at the floor, for $m = 1, \ldots, N$. D4 requires $f(\mathbf{v}_{m+1}) \leq f(\mathbf{v}_m)$ for $m = 1, \ldots, N-1$, and $f(\mathbf{v}_1) > f(\mathbf{v}_N)$. These two conditions define what we take as the correct direction of a score: on the same panel, binding equally strongly to more targets must not make a compound look more selective, and binding to one target must be more selective than binding equally strongly to all. Strict decrease at every step is not required: $R_b(k)$ with $1 \leq k < N$ equals $a - \max(x_0, b)$ for $m \leq k$ and 0 for $m > k$.

When floor-coded pairs have zero weight, the profiles $\mathbf{v}_m$ become both broader and more evenly weighted as m increases. When they keep a positive weight, they form a background of small weights, and turning them into equally strong targets moves weight from this background into a few large weights. The distribution can then become less even and the score rise, so a score can fail D4 because of its floor treatment, and each implementation is assessed with its own (example in Supporting Information, Section S4).

D4 constrains only the profiles $\mathbf{v}_m$. On a profile with unequal affinities, adding an active target that binds much more strongly than the others moves most of the weight to it, and a score that measures how concentrated binding is, such as the entropy or $P_{\max}$, then rates the compound as more selective (Supporting Information, Section S4). Whether that rating is correct depends on whether selectivity is taken to mean breadth (how many targets are bound) or concentration (how much of the binding falls on a few targets), which D4 leaves open.

We state no criterion for the ratio's choice of reference off-target, because different references answer different questions.

The Gated Entropy

For a profile with $\max_i x_i > \tau^*$ (undefined otherwise, as D1 requires), define

$$w_i = \mathbf{1}[x_i > g]\, e^{(x_i - g)/T}, \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

with the gate $g = x_0$ and the scale $T = \tau^* - x_0$, the readout difference over which a weight grows by a factor of e (1 pK unit on Davis and Klaeger, 2 on Metz, and 30 percentage points on Anastassiadis). It is the negative Shannon entropy of a gated, exponentially weighted profile. The gate gives floor-coded pairs zero weight, and a kinase just above the gate has weight close to 1. Under the entropy's hard cutoff $\max(x_i - \beta, 0)$, such a kinase has weight close to 0. On pK data the weights are powers of the association constants, $w_i \propto K_{a,i}^{1/(T \ln 10)}$. The published entropy with floor-coded pairs given zero weight has the same form with $T = 1/\ln 10 \approx 0.43$ pK units. Moving the gate without passing a value multiplies every weight above it by the same factor, which cancels in the shares, so the gated entropy satisfies D2. A reference value subtracted from every readout cancels in the same way, so the score has no baseline parameter. When the gate passes an observed value, the score jumps, so in the reversal counts the gate is evaluated at each observed value in B and just below each, as

well as at the 201 equally spaced values.

Proposition 1 (monotonicity). Adding a kinase with a value in $(x_0, \tau^*]$ never raises f if and only if $T \geq \tau^* - g$. With $T = \tau^* - x_0$, this holds for every gate $g \in B$.

The proof (Supporting Information, Section S4) has two parts. For the *if* part, every gated target has weight above 1, and the added target has weight 0 (at or below the gate) or at most $e^{(\tau^*-g)/T} \leq e$. If every weight is at least 1, appending one weight no larger than e cannot lower the entropy. For the *only if* part, for any smaller T, a profile with one active target and many kinases just above the gate scores higher after a kinase is added at τ^*.

Proposition 1 leaves T free above $\tau^* - x_0$. Two targets whose readouts differ by Δx have weights in the ratio $e^{\Delta x/T}$, so a smaller T makes the score depend more on affinity differences. We take the smallest T that D3 allows, $T = \tau^* - x_0$. This is a choice: the criteria allow any larger T. Results ("Scale of the Gated Entropy and Recovery") compares $T = 2(\tau^* - x_0)$.

Because a target at the gate has weight 0 and any target above it has weight above 1, the score jumps when a target's value crosses x_0. For example, on Davis or Klaeger ($x_0 = 5$, $T = 1$), take a profile with one active target at $\mathrm{p}K_d = 7$. Adding a second target at the floor leaves the score at 0, and adding one just above the floor lowers it by 0.53 bits. Measurement noise near the floor could therefore change scores. To measure this effect, we perturbed the measured values above the floor (normal noise with standard deviation 0.1 and 0.3 pK units), set values pushed below the floor to the floor, and left floor-coded pairs fixed. On Anastassiadis, every entry was perturbed, with standard deviation 3 and 9 percentage points. Each setting used 50 draws. Noise can also change which compounds are scored: a compound whose strongest target lies just above τ^* can fall below it, and the reverse. D1 was therefore applied to each noisy profile, and rankings were compared over the compounds scored both with and without noise.

Proposition 2 (characterization). Let a score give each target with a value above the gate g the weight $h(x_i - g)$, where h is positive, continuous, nondecreasing and the same for every target, and give the other targets zero weight. Let the score have the form $-F(\mathbf{p})$ of

the weight shares $\mathbf{p}$, where F is continuous and has Shannon's grouping property[14]: when the kinases are split into groups, F equals its value across the groups plus the share-weighted values within them. If the score satisfies D2 and D4, then h is constant or $h(t) = Ce^{t/T}$ for some $C > 0$ and $T > 0$, and the score is a positive multiple of the gated entropy with that T (with equal weights if h is constant).

The proof (Supporting Information, Section S4) has two steps. D4 gives the condition on equal shares in Shannon's theorem, which then fixes F up to a positive factor. D2, applied to profiles with two targets above τ^*, then requires the ratio of their weights to stay the same when the gate moves without passing a value, so the ratio can depend only on the difference between their readouts. Only exponential and constant h have this property, and D2 rejects every other function of the distance above the gate. For example, the entropy's weights $\max(x_i - \beta, 0)$ have this form with β as the gate: for two targets 1.0 and 0.5 units above β, lowering β by 0.5 changes their weight ratio from 2 to 1.5. The assumption that a weight depends only on the distance above the gate is needed: weights measured from a separate fixed reference satisfy D2 for any h. Rényi entropies of order other than 1 lack the grouping property, so they are excluded. By Propositions 1 and 2, the scores of this form that also satisfy D3 are the gated entropies with $T \geq \tau^* - x_0$, together with the score that weights every target above the gate equally (the limit $T \to \infty$).

We call shares $\mathbf{q}$ *more even* than shares $\mathbf{p}$ on the same panel if, for every j, the j largest entries of $\mathbf{q}$ sum to no more than the j largest entries of $\mathbf{p}$.

Proposition 3 (broadening). On a fixed panel, let a profile have positive weights $w_1, \ldots, w_r$ ($r \geq 1$) and zero weights elsewhere, and let a score be a function f of the profile's weight shares with $f(\mathbf{q}) \leq f(\mathbf{p})$ whenever $\mathbf{q}$ is more even than $\mathbf{p}$. Changing one zero weight to a with $0 < a \leq \min_i w_i$ does not raise f.

The proof (Supporting Information, Section S4) shows that the added weight, being the smallest, makes the new shares more even than the old ones. The negative Shannon and Rényi entropies, the largest share and the Gini coefficient all have this property. For a score

whose weight increases with affinity, such as the entropy, the gated entropy, $P_{\max}$ and the Gini, Proposition 3 therefore means that turning a floor-coded pair into a target bound no more strongly than the weakest weighted target never makes the compound look more selective. The condition on a is needed: adding a target tied with the strongest can lower the entropy.

Recovery from Smaller Panels

Profiling panels range from tens to hundreds of kinases, so a score computed on a smaller panel is only useful if it ranks compounds as it would on a larger one. We therefore measure how much of each score's full-panel ranking random smaller panels retain, and for how many compounds.

Subpanels of m targets were drawn uniformly at random from the curated targets, 50 per size, at $m = 10$, 20, 30, 40, 50, 65, 80, 100, 125, 150, 200, 250 and 300 below the full panel. Every score in Table 2, including the comparators, was computed on the same subpanels, with $\tau = \tau^*$ for the S-score, $\beta = x_0$ for the entropy and Gini, and $k = 1$ and $b = x_0$ for the ratio. D1 was applied to every score on every subpanel. *Coverage* is the share of compounds scored on the full panel that are still scored on a subpanel, averaged over the 50 subpanels of each size. It varies between subpanels, but on any one subpanel it is the same for every score, because D1 scores a compound under all scores or none.

Recoverability is the share of clearly separated full-panel pairs (ranks at least a quarter of the scored compounds apart) that are scored on the subpanel in the same order, with a pair tied on the subpanel counted as one half. *Conditional agreement* is the Spearman correlation between subpanel and full-panel scores over the compounds scored on both. Each score is compared with its own full-panel ranking, and its clearly separated pairs are chosen from that ranking, so recoverability measures how much of the score's own ordering a subpanel keeps. We write p^* for the smallest tested size at which the mean conditional agreement over subpanels exceeds 0.90, a descriptive benchmark that we chose. To test whether compounds

whose score depends heavily on one target are harder to recover, we compute two quantities for each compound: its rank error at 50 targets, the mean absolute change in its rank (as a share of the compounds) over the subpanels on which it is scored, and the largest change in its full-panel score when one target is left out. We report the Spearman correlation between them over the scored compounds. As published, none of the four definitions has an eligibility rule, so Supporting Information (Section S5) also reports recovery with every compound scored, without D1. These results depend partly on the conventions for undefined scores. For example, the Gini recovers as fast as the entropy once the compounds with an undefined Gini, which the convention ranks least selective, are left out.

Results

Analysis Choices Change Agreement Between Scores

With the entropy and Gini in their log-affinity forms, and compounds with an active kinase ranked among themselves, the S-score, entropy and Gini agree more with one another than with the ratio (Table 3). On Davis, where every compound has an active kinase, they correlate at 0.963–0.999, and the ratio with each at 0.51–0.56. Among the 200 Klaeger compounds with an active kinase, they correlate at 0.89–0.997, and the ratio with each at 0.38–0.64. With log-affinity weights, the entropy and Gini, like the S-score, summarize how activity is spread over the panel, while the ratio compares the primary target with one off-target.

Weighting. The grouping of the S-score, entropy and Gini holds only for the log-affinity forms of the entropy and Gini. The published entropy, with association-constant weights, correlates more with the ratio than with the S-score or Gini on Davis, Klaeger and Metz, under both treatments of floor-coded pairs (Table 3 and Supporting Information, Table S2). On Anastassiadis, the native percent-inhibition Gini correlates with the entropy at 0.63, and with the S-score and the ratio at only 0.34 and 0.33.

Table 3: Spearman correlations between median rankings. S is the S-score, E the entropy, G the Gini and R the ratio, with the entropy and Gini in their log-affinity forms. E_{K_a}: the published entropy, with floor-coded pairs given zero weight and, in parentheses, kept at the floor. In the active columns, compounds are ranked among themselves. Every Davis compound has an active kinase.

	Davis	Klaeger	
Pair	all ($n = 68$)	all ($n = 222$)	active ($n = 200$)
S–E	0.963	0.896	0.897
S–G	0.963	0.508	0.892
E–G	0.999	0.658	0.997
S–R	0.563	0.078	0.378
E–R	0.513	0.344	0.622
G–R	0.510	0.655	0.635
E_{K_a}–S	0.738 (0.344)	0.571 (−0.597)	0.564 (−0.412)
E_{K_a}–G	0.736 (0.368)	0.530 (0.077)	0.782 (−0.094)
E_{K_a}–R	0.861 (0.688)	0.657 (0.648)	0.916 (0.591)

How far the published entropy is from the S-score and Gini depends on the treatment of floor-coded pairs. They make up 71% of the Davis matrix, and the published entropy's correlation with the S-score there is 0.34 with them at the floor and 0.74 with them given zero weight. The published and log-affinity entropies correlate at 0.37 (Davis) and −0.31 (Klaeger) with floor-coded pairs at the floor, and at 0.74 and 0.79 with them given zero weight (Figure 1). The negative Klaeger value comes from the floor-coded pairs, with which the published entropy also fails D4 ("Criteria and the Gated Entropy"). Using every measured Klaeger affinity at its measured value, including the 194 weaker than the floor, changes the published entropy's ranking little (Spearman 0.9996 with the floor-excluded ranking among active compounds) and changes its correlations and recovery by at most 0.02 (Supporting Information, Section S2).

Compounds ranked. Twenty-two Klaeger compounds have no target above the activity threshold. With them included, the Gini correlates more with the ratio (0.66) than with the S-score (0.51), although the difference is not significant (bootstrap 95% interval of the

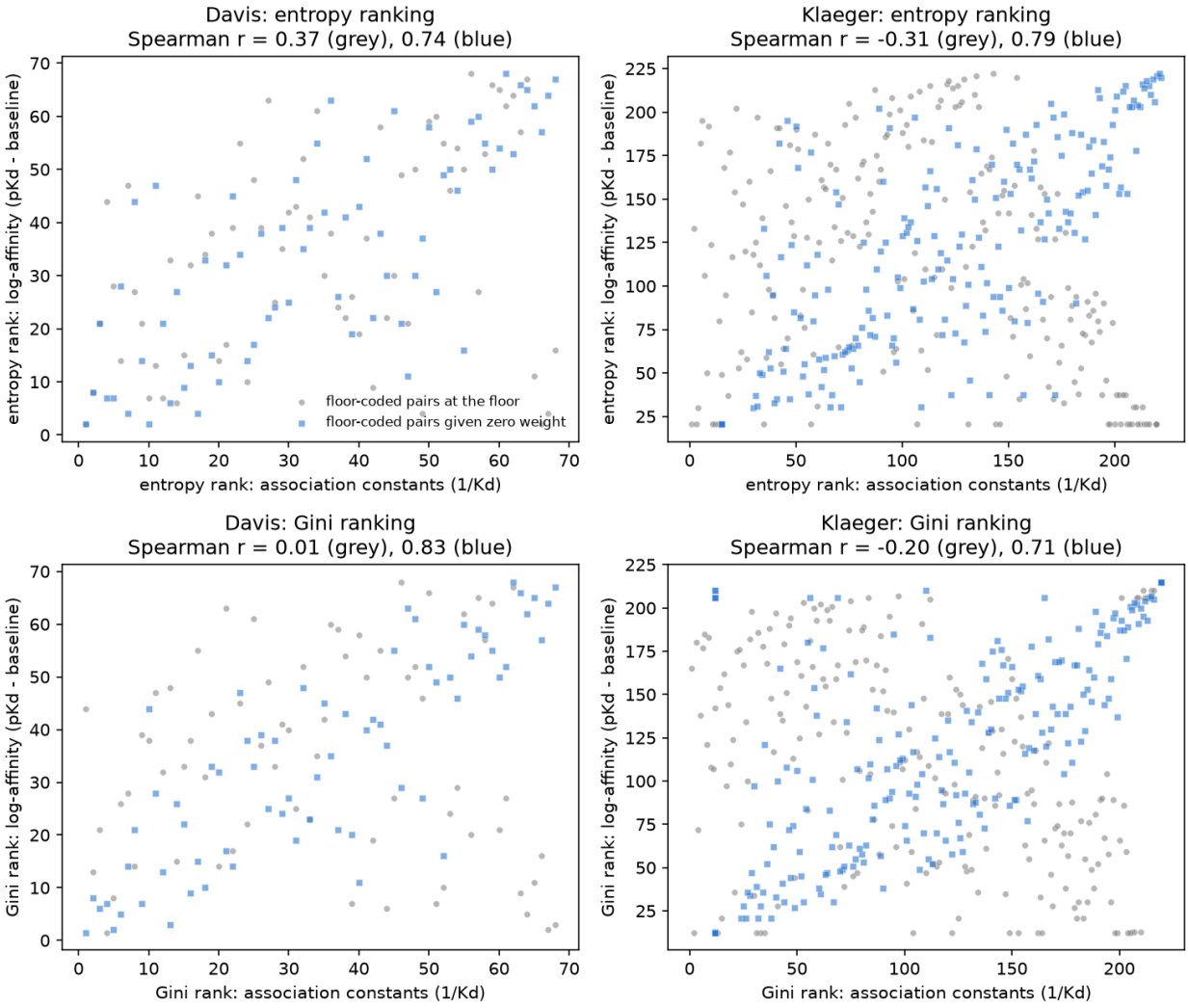


Figure 1: Rankings under association-constant weights (x-axis) against the baseline-subtracted log-affinity form (y-axis, median rank over the baseline sweep), on Davis and Klaeger. Top: entropy, for which association-constant weights are the published form. Bottom: Gini, shown for comparison. Its published form uses percent inhibition. Grey: floor-coded pairs kept at the floor. Blue: floor-coded pairs given zero weight. Spearman r above each panel.

difference $[-0.35, 0.06]$, against $[0.16, 0.36]$ among active compounds). Each definition scores these compounds by its own convention. The S-score and entropy rank these 22 compounds among the most selective (median ranks 24 and 20 of 222) and the ratio near the bottom (182). The Gini ranks them near the top where its score is defined and near the bottom where it is not (mean ranks 32 and 212). Their aggregated rank instability is $\sigma = 77.9$, against 32.2 for active compounds ($p < 0.001$, Mann–Whitney U test). Within a single definition's sweep, their rank instability is significantly higher than that of active compounds only for the Gini (65.0 against 16.3). The difference reverses for the S-score (19.0 against 25.7, $p = 0.01$), whose score is 0 for all of these compounds at thresholds of 6 or above. Sixteen of the 22 bind some kinase above the floor. Computed only over the configurations where their scores are defined, the mean instability of these 16 is lower than that of the active compounds within the entropy and the Gini (7.0 against 13.5, and 6.7 against 13.6, both with $p = 0.002$). The other six show no binding to any curated target, although all are established inhibitors. For three of them, the target was quantified for other compounds (Supporting Information, Section S3). Whether a compound is eligible for a selectivity score therefore depends on the panel and the threshold. Supporting Information (Section S3) gives the per-definition contrasts.

Agreement within each dataset. Among compounds with an active kinase, the S-score, entropy and Gini correlate at 0.81–0.999 on all four datasets, and the ratio less with each (Supporting Information, Figure S1). On Anastassiadis, the separation is weak: the ratio correlates with the other three at 0.79–0.84, against 0.87–0.999 among them. With all compounds included, this falls to 0.48–0.55, because of the 45 compounds with no active kinase. On Metz, the ratio correlates with the other three at only 0.15–0.20, because the other three largely follow how many kinases each compound was tested against. Untested pairs are coded at the floor, and nearly every tested value (99.9%) lies above it. A compound tested against more kinases therefore has more kinases with positive weight and looks less selective under the S-score, entropy and Gini, while the ratio, which uses only the strongest value and

one of the next five, is barely affected. The Spearman correlation between a compound's coverage and its median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score and 0.11 for the ratio. Compounds measured against more kinases are also active against a larger share of those measured, but that association is much weaker ($r = 0.33$). The Metz correlations therefore show how coding untested pairs at the floor affects rankings.

Target selection and filtering. On the released Davis panel, the ratio agrees less with the other definitions than on the curated panel (0.34–0.46 against 0.51–0.56, Supporting Information, Section S1). Restoring the low-confidence Klaeger pairs shifts the all-compound Klaeger correlations by at most 0.08 and the active-compound correlations by at most 0.01. It leaves every p^* of the recovery analysis unchanged (Supporting Information, Section S1).

Effects of scoring against a different dataset versus a different measure. On the 46 compounds profiled in both Davis and Klaeger, the same definition's Davis and Klaeger rankings correlate, on the 201 kinases both panels share, at 0.34–0.84 (mean 0.61), and different definitions' rankings within one dataset at −0.08–1.00 (mean 0.57). Among the 40 compounds active in both datasets, the means are 0.72 and 0.73, and the Gini's cross-dataset correlation rises from 0.45 to 0.84. On these compounds, the choice of definition moves rankings about as much as the change of dataset. The 46 compounds are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average, and without restricting the panels to their shared kinases the cross-dataset correlations are lower (mean 0.53).

Parameter Sensitivity and Profile Associations

Moving each definition's activity boundary across B reorders a substantial share of compound pairs. The S-score's threshold reverses 8.4–13.0% of pairs of scored compounds on Davis, Klaeger and Anastassiadis, and 21.8% on Metz. The entropy's and Gini's baselines reverse 7.9–12.4% on the same three datasets, and 22.5–22.8% on Metz. The ratio's level b, which

replaces any off-target value below b, measured or not, reverses 3.0–13.7% on Klaeger and 2.1–9.7% on Anastassiadis, rising with k, 0–2.6% on Davis and 0.04–1.03% on Metz. Sweeping the gated entropy's gate, with T fixed at $\tau^* - x_0$, reverses 9.2–14.6% of pairs on Davis, Klaeger and Anastassiadis, and 23.8% on Metz. On Klaeger, the per-compound instability under the gate sweep is close to that under the entropy's baseline sweep (mean rank $\sigma = 6.7$ and 7.0, Supporting Information, Figure S2). Supporting Information (Table S3) gives every value.

Reference off-target. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the first kinase with the second, treating the second as an off-target (a kinase the compound is not meant to bind). For near-tied kinases this difference is close to zero, so the compound scores as non-selective. $R(2)$ compares the first kinase with the third, treating the second as a co-target (a second intended target), so the same compound can score as highly selective. On the 22 Klaeger compounds with an active kinase and a top_1–top_2 gap below 0.05, mean $R(1) = 0.02$ against $R(2) = 0.50$. Which reference is appropriate depends on whether the second kinase is an intended target, which the profile does not show. On Davis, fedratinib, a JAK2 inhibitor with JAK2 and GAK tied, has the most unstable ratio ranking (Figure 2, left).

Profile associations. Ranked among compounds with an active kinase, three associations between profile features and instability are significant at $p < 0.05$ on both pK_d datasets (Supporting Information, Table S4). The number of active kinases correlates negatively with entropy instability on every dataset ($r = -0.30$ to -0.42, on Davis only nominally, with $p = 0.01$), and with Gini instability on Davis and Klaeger ($r = -0.31$ and -0.25). On Klaeger, entropy instability correlates with the share of a compound's above-floor kinases that lie within the baseline sweep ($r = 0.22$, $p = 0.002$). The top_1–top_2 gap correlates with ratio instability on Davis, Klaeger and Metz ($r = -0.44$, -0.28 and -0.26, all $p < 0.001$), but not on Anastassiadis ($r = 0.00$, or -0.15 with $p = 0.10$ without eight compounds whose two best targets are at the 100% inhibition ceiling). With a Bonferroni correction, only the

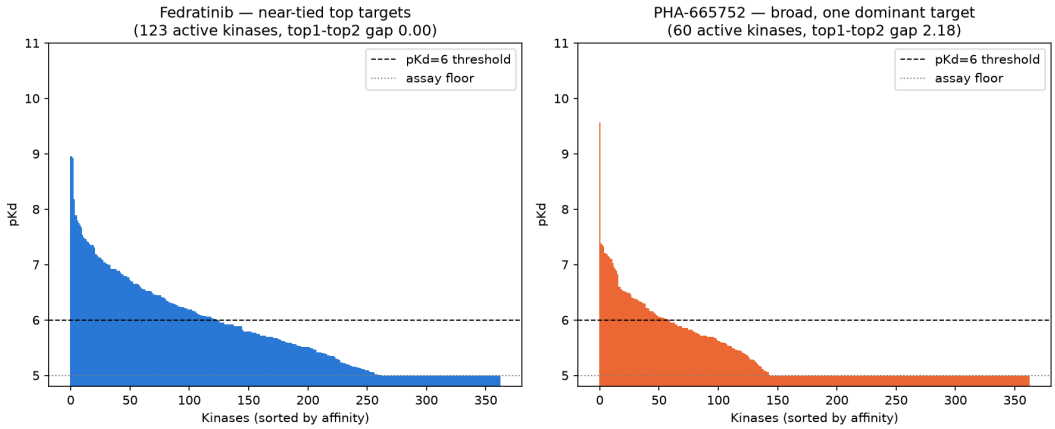


Figure 2: Binding profiles of two Davis compounds, targets sorted by affinity. Left: fedratinib, the most ratio-sensitive compound, with JAK2 and GAK tied among 123 active targets. Right: PHA-665752, the most unstable compound overall ($\sigma = 16.4$), with one dominant target among 60 active targets. Dashed line: p$K_d = 6$ activity threshold. Dotted line: p$K_d = 5$ assay floor.

ratio–gap association remains significant on both pK_d datasets, as it does with the Holm and Benjamini–Hochberg procedures. On Klaeger (but not on Davis), S-score instability also correlates with the range of active affinities ($r = -0.32$, and partial $r = -0.22$ with the number of active kinases held fixed). Ranked among all 222 Klaeger compounds, the skewness of the active affinities was the strongest correlate of Gini instability ($r = +0.26$). Ranked among active compounds, it is not significant ($r = +0.07$, $p = 0.31$), so the marginal association came mainly from the scoring of compounds with no active kinase. With the number of active kinases held fixed, a weaker association remains (Supporting Information, Section S3).

Broad profiles. Among active compounds, seven of the ten largest entropy–ratio disagreements on Klaeger involve profiles with 14–27 active kinases, and six of these have one clearly dominant target (top_1–top_2 gap above 1). The ratio ranks these seven 31st to 63rd of 200 and the entropy and Gini 146th to 180th. This is because the ratio reflects the dominant target, and the entropy and Gini reflect the number of active kinases. Bosc et al.[8] attributed disagreements for individual compounds to such profiles (i.e. broad profiles with one dominant

target). On Davis, the most unstable compound overall is a profile of this kind (Figure 2, right).

Criteria and the Gated Entropy

Table 4 gives each score's verdict on the four criteria, D1 to D4, all judged by construction (Methods). None of the four definitions, as published, withholds the score of a compound with no active kinase (so none meets D1), but any score can meet it by adopting that rule. The comparators and the gated entropy adopt it. D2 and D3 are the criteria on which the scores differ. The entropy, the Gini and the ratio violate D2, and the other scores satisfy it. The published entropy and $P_{\max}$ have no boundary parameter, so D2 does not apply to them. The S-score with $\tau > x_0$, the entropy with $\beta = x_0$, the Gini and the published entropy violate D3. The other scores, including $P_{\max}$, satisfy it. Every score in the table satisfies D4, and so do $P_{\max}$ and the published entropy with floor-coded pairs given zero weight. D2 does not make a ranking independent of where the boundary is placed: for every score, moving the boundary across B reverses at least 13.7% of pairs on at least one dataset (Supporting Information, Table S3).

Table 4: The four criteria for the four definitions, three comparators and the gated entropy. $P_{\max}$ and the published entropy have no boundary parameter and are assessed in the text. With $\beta < x_0$, the entropy and Gini fail D4. ✓: satisfied. ×: violated.

	S-score		Count	Entropy		Gini	Ratio	Gated
	$\tau > x_0$	$\tau = x_0$		$\beta = x_0$	$\beta = \tau^*$	$\beta \geq x_0$		entropy
D1 (eligibility)	×	✓	✓	×	✓	×	×	✓
D2 (locality)	✓	✓	✓	×	×	×	×	✓
D3 (monotonicity)	×	✓	✓	×	✓	×	✓	✓
D4 (orientation)	✓	✓	✓	✓	✓	✓	✓	✓

By construction. These verdicts follow from each score's definition, without data. The S-score and the active count depend on the boundary only through which targets lie above it, so they satisfy D2. The entropy and Gini subtract β from every readout above it, and $R_b(k)$ replaces a reference off-target below b by b. Moving the boundary therefore changes

these scores even when it passes no value, and they violate D2. Moving the gated entropy's gate multiplies every weight above it by the same factor, so it satisfies D2. The S-score violates D3 for any threshold above the floor, because adding a kinase bound between the floor and the threshold increases N but not the number of hits, which makes the compound look more selective. With the threshold at the floor, every addition that D3 allows is a hit, and a compound with C hits among N kinases moves from $-C/N$ to $-(C+1)/(N+1)$, which is not higher. The active count ranks compounds on any one panel as the S-score at τ^* does, but it does not divide by N. A kinase added at or below τ^* is not counted, so the count and the score stay the same, and D3 holds. D3 therefore concerns how scores change when a panel is extended. The Gini violates D3, since a weakly bound kinase can raise its inequality. The ratio satisfies it, because a weak addition cannot displace the primary target or lower the reference off-target. $P_{\max}$ satisfies it too, since a weak addition leaves its numerator unchanged and raises its denominator. The entropy satisfies D3 at $\beta = \tau^*$, where weak binding carries zero weight. The gated entropy satisfies D1 through its eligibility rule and D3 by Proposition 1 in Methods. D3 concerns adding a target, not strengthening one already in the profile (so strengthening one is allowed to raise the gated entropy's score). For example, with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1, the score is -5.502. Raising one of the fifty to 6.0 raises it to -5.491.

D4 asks that a score not rise when a compound binds one more kinase as strongly as the others. For any compound that binds m kinases at the same value a, and the rest of the panel at the floor (such profiles are denoted by $\mathbf{v}_m$ in Methods), the scores are:

- active count: $-m$
- S-score with $\tau < a$: $-m/N$
- the entropies and the gated entropy, whose m weights are equal: $-\log_2 m$
- $P_{\max}$: $1/m$
- Gini: $(N-m)/N$

- ratio $R_b(k)$: $a - \max(x_0, b)$ for $m \leq k$ and 0 otherwise.

None of these values rises as m grows, so all of these scores satisfy D4. The values assume that floor-coded pairs have zero weight, which, for the entropy and Gini, requires $\beta \geq x_0$. On Anastassiadis, their sweeps also include baselines below the 20% floor, where both fail D4: on its 291 kinases, with $\beta = 0\%$ and targets at 60%, a second target raises the Gini from 0.0068 to 0.0135. With floor-coded pairs kept at the floor, the published entropy also fails D4 on panels of the datasets' sizes: on 363 kinases with targets at $\mathrm{p}K_d = 7$, a second target raises its score from -7.41 to -6.76 bits. $P_{\max}$ satisfies D4 under either treatment, because its largest share falls as m grows.

D1 to D3 alone do not fix the direction of a score. The active count with its sign reversed, which calls a compound more selective the more active kinases it has, depends only on which targets lie above τ^*, and a kinase added at or below τ^* leaves it unchanged. It therefore satisfies D1 to D3, as the gated entropy does, but it rises with the number of equally strong targets ($+m$ on $\mathbf{v}_m$), contradicting D4. The gated entropy without its non-binder gate also fails D4, because floor-coded pairs carry weight, and adding equally strong targets can concentrate the distribution. For the entropy, the gated entropy, $P_{\max}$ and the Gini, Proposition 3 gives a conclusion stronger than D4. On any profile (not only one with equally strong targets), turning a floor-coded pair into a target that is no stronger than the weakest weighted one does not raise the score.

By construction: counterexamples. The published entropy violates D3: with floor-coded pairs given zero weight, a profile with one target at $\mathrm{p}K_d = 6.01$ and fifty at 5.01 scores -5.35 bits, and adding one at 6.0 raises this to -5.19. The entropy violates D3 at $\beta < \tau^*$. Its hard cutoff gives a kinase just above β a weight close to zero. A profile with many such kinases therefore has high entropy. Adding one kinase bound more strongly, but still below τ^*, concentrates the distribution. With $\beta = 5.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 scores -4.89 bits, and adding one at 6.0 raises this to -4.83. The gated entropy does not violate D3 on this profile: adding the kinase at 6.0 lowers its score from -5.502

to -5.522, because the exponential weights give that kinase only 2.5 times the weight of a kinase at 5.1 (2.72 against 1.11). The entropy's hard cutoff gives it 10 times the weight (1.0 against 0.1).

Empirical checks. To additionally check empirically that Proposition 1 holds, we tested D3 for the entropy and the gated entropy on the measured profiles. On the measured profiles of the active compounds, adding a hypothetical kinase at τ^* never raises the entropy's score on Davis, Klaeger or Metz at any baseline in their sweeps. On Anastassiadis, at baselines below the 20% floor (33% of the compounds at $\beta = 0\%$, 16% at 5%), it does. It never raises the gated entropy's score on any dataset. Adding measured kinases bound below the threshold to random 50- or 100-kinase subpanels of the same profiles raises the entropy's score in at most 0.1% of cases on any dataset, and never the gated entropy's.

On the four datasets, we compared the gated entropy's rankings with those of other scores, and tested their sensitivity to τ^* and to noise. On the data, the gated entropy's score falls as the number of active kinases rises (Spearman -0.76 to -0.95). Without the non-binder gate, the correlation reverses ($r = +0.67$ on Klaeger). Its rankings agree with the entropy's (0.969–0.993) and with those of the S-score with $\tau = x_0$ (0.964–0.996), which also satisfies D1 to D4. Moving τ^* from 5.5 to 7 (40% to 70% on Anastassiadis), with $T = \tau^* - x_0$ moved with it, changes which compounds are scored and how they are ranked: over the compounds scored at both values, the Spearman correlation with the default ranking is 0.85 to 0.999. The lowest values are at $\tau^* = 5.5$ on Davis and Klaeger, where T falls to 0.5 pK units. With random noise added to the measured values, as described in Methods, D1 changed the status of at most 11 compounds in any draw. Over the compounds scored both with and without noise, the gated entropy's ranking correlated with its noise-free ranking at Spearman 0.972 or higher, and the entropy's at 0.983 or higher. On these datasets, the gated entropy is not an empirical improvement on simpler scores. Its difference from the entropy is in construction: the entropy satisfies D3 at $\beta = \tau^*$, where binding below the threshold has zero weight, and the gated entropy satisfies D3 while giving that binding weight.

What the criteria determine. The active count, the S-score with $\tau = x_0$ and the gated entropy satisfy all four criteria, and so does every gated entropy with $T \geq \tau^* - x_0$. Proposition 2 narrows the choice within one class of scores: those that weight each target above the gate by the same continuous, nondecreasing function of its distance above the gate and combine the weight shares with Shannon's grouping property. In this class, D2 and D4 reject everything but the gated entropies and the score that gives every target above the gate the same weight, $-\log_2 n$ for n such targets (the limit $T \to \infty$). D3 then additionally retains only the gated entropies with $T \geq \tau^* - x_0$ (Proposition 1) and the equal-weight score. With the gate and T fixed, the score is determined up to a positive multiple. Taking the smallest admissible T is our choice. On Klaeger, the rankings for $T = 1$, 1.5 and 2 agree at Spearman 0.985 or higher, and other variants are compared in Supporting Information, Section S4.

Recovery from Small Panels

On small random panels, some compounds can no longer be scored. At 50 targets, 88% of the compounds scored on the full panel are still scored on Davis, 71% on Klaeger, 77% on Anastassiadis and 99% on Metz (Figure 3, top). The compounds lost are mostly those with few active targets: on a 50-target Klaeger subpanel, a compound with one active target stays scored with probability 0.23, and one with ten with probability 0.93. High conditional agreement on a small panel can therefore coexist with the loss of many narrowly active compounds. Coverage limits the recovery of pair order. Keeping 90% of clearly separated pairs takes 150 targets on Davis, 200 on Klaeger and Anastassiadis, and 40 on Metz for every score except the ratio, the published entropy and $P_{\max}$, which need as many or more (Figure 3, middle).

Among compounds scored on both panels, scores whose ranking depends mostly on the few strongest targets recover slowest (Figure 3, bottom). The S-score with $\tau = x_0$, the entropy with $\beta = x_0$ and the Gini rank compounds almost as the share of targets with detected binding does (Spearman above 0.97 with the S-score at $\tau = x_0$ on every dataset). We call

these three the *share-following scores*. The curves of the entropy and Gini differ from the S-score's by at most 0.012, so Figure 3 draws only the S-score with $\tau = x_0$. They recover fastest on Davis, Klaeger and Anastassiadis, reaching p^* at 20, 50 and 30 targets, against 30, 80 and 50 for the S-score at τ^* and the active count (which rank compounds identically on any one panel and share one curve in Figure 3), and 40, 100 and 65 for the entropy with $\beta = \tau^*$. On Metz, where untested targets are set to the floor, the S-score at τ^*, the active count and the entropy with $\beta = \tau^*$ recover fastest (50 targets, against 65 for the share-following scores). With bars of 0.80, 0.85 or 0.95 instead of 0.90, these fastest scores still reach the bar no later than any score outside their group. The published entropy and $P_{\max}$, with floor-coded pairs given zero weight, need 300 of 363 targets on Davis and 200 of 217 on Klaeger. On Metz, the published entropy needs 150 of 172, while $P_{\max}$ needs more than the largest subpanel tested (150). The ratio needs 200 of 217 targets on Klaeger and more than the largest subpanel tested on the other datasets (300 of 363, 250 of 291 and 150 of 172). The published entropy gives weak binding a small positive weight, so including weak binding does not by itself make a score recover quickly. The gated entropy follows the share of targets with detected binding less closely (Spearman 0.964–0.996 with the S-score at $\tau = x_0$). Its mean Spearman curve lies below that of the S-score with $\tau = x_0$ on every dataset, by up to 0.069, and it reaches p^* at 50, 65, 30 and 80 targets on Davis, Klaeger, Anastassiadis and Metz.

The value p^* is a threshold on the mean: at p^*, 62–90% of individual subpanels exceed 0.90 for the share-following scores (Supporting Information, Section S5). On Davis, Anastassiadis and Metz, compounds whose score changes most when one target is left out tend to have larger rank errors at 50 targets (Spearman $r = 0.18$–0.70 for the entropy with $\beta = x_0$, the Gini and the gated entropy). On Klaeger, this association is weak ($r \leq 0.13$).

Scale of the Gated Entropy and Recovery

The gated entropy satisfies D1 to D4 for any scale $T \geq \tau^* - x_0$. With $T = 2(\tau^* - x_0)$, it reaches p^* at 20, 50, 30 and 65 targets on Davis, Klaeger, Anastassiadis and Metz, the values

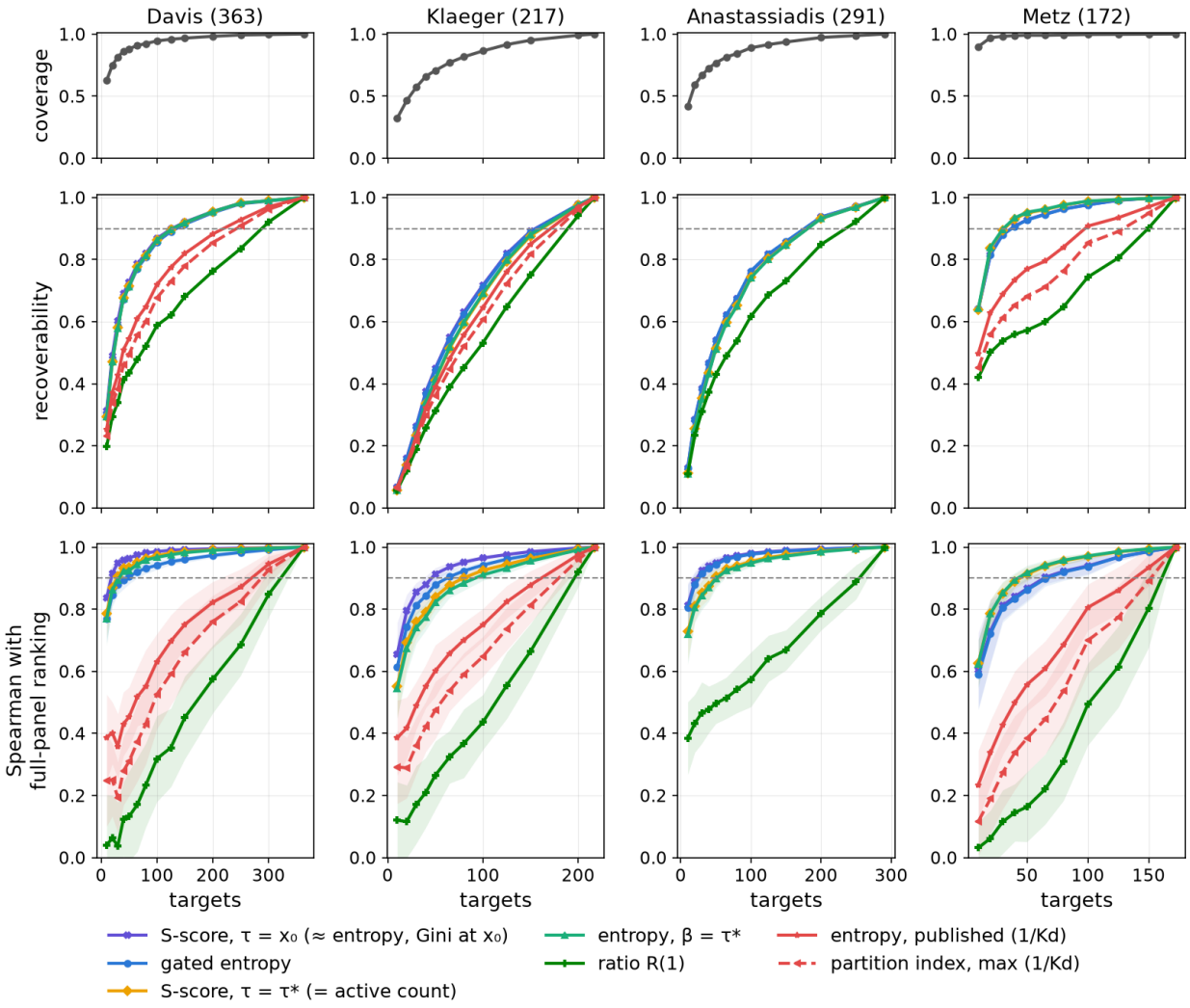


Figure 3: Recovery from random subpanels with D1 applied to every score, on all four datasets (50 subpanels per size). Top: coverage, the share of compounds scored on the full panel that are still scored. Middle: recoverability, the share of clearly separated full-panel pairs that are scored on the subpanel and keep their order. Bottom: mean Spearman correlation between subpanel and full-panel rankings over compounds scored on both, with ±1 standard deviation across subpanels shaded. Each score is compared with its own full-panel ranking. The published entropy and $P_{\max}$ give floor-coded pairs zero weight and are not defined on Anastassiadis. Dashed line: 0.90.

of the share-following scores, against 50, 65, 30 and 80 with $T = \tau^* - x_0$. A larger scale gives the targets more nearly equal weights, and the score then ranks compounds almost as the S-score with $\tau = x_0$ does (Spearman 0.9968–0.9996). Recovery compares each score with its own full-panel ranking, so the faster recovery does not show that the score with the larger scale measures selectivity better. The other analyses keep $T = \tau^* - x_0$, the admissible scale at which the score depends most on affinity differences. Supporting Information, Section S5, gives the same comparison for $T = 4(\tau^* - x_0)$.

Discussion

The scores compared here summarize different aspects of a binding profile, and the discrepancies between their rankings depend on choices made before any score is computed. The S-score, entropy and Gini agree more with one another than with the ratio when the entropy and Gini are computed from readouts above a common baseline and compounds without an active kinase are left out. With association-constant weights, or with compounds that have no active kinase included, this grouping weakens or disappears. When most entries are coded at the floor, as on Metz, the S-score, entropy and Gini agree for a different reason: each ranks compounds largely by how many kinases the compound was tested against. Bosc et al.'s window score correlates strongly with the ratio (0.91–0.96), and their ranking score is a ratio by definition. A proxy for the CATDS score of Klaeger et al., the top target's share of the profile, instead correlates more with the Gini than with the ratio on every dataset (Supporting Information, Section S6). The ratio and the S-score, entropy and Gini disagree most on broad profiles with one dominant target. A selectivity ranking is therefore a statement about a score, its implementation, and a set of compounds and targets. On compounds and kinases shared by two datasets, changing the definition moved rankings about as much as changing the dataset.

Roles of the Criteria and the Gated Entropy

D1 is a reporting rule that any score can adopt. D2 and D3 depend on a score's form, and they separate the scores. The entropy, Gini and ratio fail D2, and the S-score above the floor, the entropy with $\beta = x_0$, the Gini and the published entropy fail D3. D4 fixes the direction of a score, which D1 to D3 leave open. The criteria do not single out one score. Meeting D2 does not make rankings independent of the boundary, so the boundary must still be fixed by protocol. D3 concerns adding a weakly bound kinase, above the floor and at or below the activity threshold. A kinase added at the floor has zero weight in the gated entropy and leaves its score unchanged. No kinase added at or below the activity threshold therefore raises the gated entropy's score, and most entries lie in this range. A stronger value for a target already in the profile can raise the score.

Among the scores studied, only the gated entropy meets all four criteria and weights targets by affinity. Within the class of Proposition 2, the criteria leave only exponential weights, and D3 requires the scale to be at least the distance between the floor and the activity threshold. The published entropy, with floor-coded pairs given zero weight, is a member of this family whose scale is smaller than this distance on these datasets, so it violates D3. As the scale grows, the rankings approach those of the share of targets with detected binding. Two simpler scores rank compounds almost as the gated entropy does, and each lacks one of its properties. The entropy with $\beta = x_0$ weights targets by affinity but fails D2 and D3. The S-score with $\tau = x_0$ meets all four criteria but weights every detected target equally. With twice the smallest allowed scale, the gated entropy keeps its affinity weights and recovers rankings from small panels as fast as that S-score.

A score's correlation with the number of active kinases on real data does not show whether it satisfies D4, which concerns only profiles with equally strong targets. On real profiles, a compound with more active targets can have more concentrated binding, and a score that measures concentration can then rank it as more selective without violating D4.

Reporting Recommendations

Each recommendation follows from a result above.

1. State which targets the score is computed over, including how the panel was curated, and the score implementation, including the weighting and its parameters, such as a baseline or the gated entropy's scale.

2. Distinguish measured affinities, pairs with no detected binding, rejected fits and untested pairs, and state the value and weight each receives.

3. Apply D1: state the activity threshold, withhold or flag the scores of compounds with no target above it, and rank the eligible compounds among themselves.

4. Fix the activity boundary in the analysis protocol and report how rankings change when it is moved, for example by each compound's rank instability across the sweep. Rank instability is not a guide to robustness across datasets.

5. For small panels, report how many compounds remain scorable as well as how well the ranking of those compounds is recovered.

6. When comparing studies that use different scores, fix the comparison protocol in advance and report the compounds on which the scores disagree.

Limitations

The four datasets, from three assay technologies, cover 68 to 704 compounds and 172 to 363 kinases, and the results may not hold for other compounds or panels. The datasets also differ in how many interactions they find: the 46 compounds profiled on both Davis and Klaeger are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average. On Davis, Klaeger and Anastassiadis every compound was tested against the whole panel, and only 1.1% of the Anastassiadis entries are missing (set to 0% inhibition). On Metz,

each compound was tested against some particular subset of the 172 kinases. On no dataset is the floor a measured detection limit. It codes non-detection (on Davis), non-detection, low-confidence fits and weak measured affinities (on Klaeger), and untested pairs (on Metz). On Anastassiadis, it is an operational cutoff for weak inhibition. Fixing the boundary at the floor therefore depends on how each dataset codes these pairs. Random subpanels are one sampling design, so panels chosen by kinase family, as many screening panels are, may recover rankings differently. Cell-free panels can also miss interactions that cellular target-engagement assays detect[19,20].

No score considered here uses target identity or exposure. Two compounds with the same affinity distribution receive the same score whether their off-targets are inert or linked to a class toxicity, and an affinity alone does not establish whether a target is occupied at a tolerated dose. A selectivity score is at best a summary of the binding profile, to be weighed with target-specific liability knowledge and exposure modeling. Selectivity rankings of the approved Klaeger compounds were not associated with the proportions of their adverse-event reports that were serious, fatal or led to hospitalization (Supporting Information, Section S7).

Conclusion

Kinase inhibitor selectivity rankings depend on choices usually left implicit: how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which targets the panel includes, and where the activity boundary is placed. These choices change whether the common scores agree, how much a ranking moves when the activity boundary moves, and how much of a ranking a small panel retains. Selectivity analyses should state them, apply an activity-eligibility rule, fix the activity boundary in advance and report its effect, and report how many compounds remain scorable on small panels. The gated entropy gives weight to binding below the activity threshold and remains monotone under weak-target addition. Among scores that weight each target by its distance above

the activity boundary and satisfy Shannon's grouping rule, the criteria leave only the gated entropies. Their scale is the readout difference over which a weight grows by a factor of e. It must be at least the distance between the floor and the activity threshold. The smallest scale makes the score depend most on affinity differences. On these data, it recovers rankings from small panels more slowly than the share of targets with detected binding. Twice the scale gives the targets more nearly equal weights, and the score then recovers rankings as fast, and also ranks compounds almost as that share does. Analyses that use the gated entropy should therefore report its scale.

Acknowledgement

The author thanks the developers of the Davis, Klaeger, Anastassiadis, and Metz datasets for making their data publicly available. The author used Claude (Anthropic, claude.ai), a large language model assistant, during the preparation of this manuscript. Specifically, Claude was used to assist with: data analysis code development and debugging in Python; literature search and identification of related work; drafting and editing of manuscript text; and research planning and interpretation of results. All scientific conclusions, the conceptual framework, the criteria D1–D4, and the final manuscript content were developed, verified, and approved by the author. The author takes full responsibility for the accuracy and integrity of all content in this work. The use of Claude does not transfer any rights over the content of this manuscript to Anthropic or any third party.

Supporting Information

Sections S1–S8 of the Supporting Information, with Tables S1–S9 and Figures S1–S4, give the full results for each analysis, the proofs of Propositions 1–3 and the script behind every reported number.

Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`; `https://doi.org/10.1038/nchembio.530`). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at `https://github.com/polinavino/kinase-selectivity-definitions`; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FAERS adverse-event report rates for the FDA-approved Klaeger compounds, obtained via openFDA queries). The analyses used Python 3 with NumPy[21], pandas[22] and SciPy[23], and Supporting Information (Table S9) names the script behind every result. No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code. A preprint of this manuscript is available on ChemRxiv at `https://doi.org/10.26434/chemrxiv.15001618/v1`.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest

The author declares no conflict of interest.

References

[1] Roskoski, R. Properties of FDA-approved small molecule protein kinase inhibitors: a 2026 update. *Pharmacol. Res.* **2026**, *224*, 108107.

[2] Manning, G.; Whyte, D. B.; Martinez, R.; Hunter, T.; Sudarsanam, S. The protein kinase complement of the human genome. *Science* **2002**, *298*, 1912–1934.

[3] Bhullar, K. S.; Lagarón, N. O.; McGowan, E. M.; Parmar, I.; Jha, A.; Hubbard, B. P.; Rupasinghe, H. P. V. Kinase-targeted cancer therapies: progress, challenges and future directions. *Mol. Cancer* **2018**, *17*, 48.

[4] Karaman, M. W.; Herrgard, S.; Treiber, D. K.; et al. A quantitative analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2008**, *26*, 127–132.

[5] Uitdehaag, J. C. M.; Zaman, G. J. R. A theoretical entropy score as a single value to express inhibitor selectivity. *BMC Bioinformatics* **2011**, *12*, 94.

[6] Graczyk, P. P. Gini coefficient: a new way to express selectivity of kinase inhibitors against a family of kinases. *J. Med. Chem.* **2007**, *50*, 5773–5779.

[7] Cheng, A. C.; Eksterowicz, J.; Geuns-Meyer, S.; Sun, Y. Analysis of kinase inhibitor selectivity using a thermodynamics-based partition index. *J. Med. Chem.* **2010**, *53*, 4502–4510.

[8] Bosc, N.; Meyer, C.; Bonnet, P. The use of novel selectivity metrics in kinase research. *BMC Bioinformatics* **2017**, *18*, 17.

[9] Bello, T.; Gujral, T. S. KInhibition: a kinase inhibitor selection portal. *iScience* **2018**, *8*, 49–53.

[10] Saifudeen, M.; Zhu, S.; Liang, S.; et al. Comprehensive profiling of clinically approved kinase inhibitors reveals mutation-specific inhibitors and opportunities for drug repurposing. *Nat. Biotechnol.* **2026**, DOI: 10.1038/s41587-026-03090-8.

[11] Miljkovic, F.; Bajorath, J. Reconciling selectivity trends from a comprehensive kinase inhibitor profiling campaign with known activity data. *ACS Omega* **2018**, *3*, 3113–3119.

[12] Klaeger, S.; Heinzlmeir, S.; Wilhelm, M.; et al. The target landscape of clinical kinase drugs. *Science* **2017**, *358*, eaan4368.

[13] Uitdehaag, J. C. M.; Verkaar, F.; Alwan, H.; de Man, J.; Buijsman, R. C.; Zaman, G. J. R. A guide to picking the most selective kinase inhibitor tool compounds for pharmacological validation of drug targets. *Br. J. Pharmacol.* **2012**, *166*, 858–876.

[14] Shannon, C. E. A mathematical theory of communication. *Bell Syst. Tech. J.* **1948**, *27*, 379–423.

[15] Davis, M. I.; Hunt, J. P.; Herrgard, S.; et al. Comprehensive analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2011**, *29*, 1046–1051.

[16] Öztürk, H.; Özgür, A.; Ozkirimli, E. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics* **2018**, *34*, i821–i829.

[17] Anastassiadis, T.; Deacon, S. W.; Devarajan, K.; Ma, H.; Peterson, J. R. Comprehensive assay of kinase catalytic activity reveals features of kinase inhibitor selectivity. *Nat. Biotechnol.* **2011**, *29*, 1039–1045.

[18] Metz, J. T.; Johnson, E. F.; Soni, N. B.; Merta, P. J.; Kifle, L.; Hajduk, P. J. Navigating the kinome. *Nat. Chem. Biol.* **2011**, *7*, 200–202.

[19] Binder, M. J.; Bashore, F. M.; Dunn Hoffman, K. K.; et al. Cellular context influences kinase inhibitor selectivity. *J. Med. Chem.* **2026**, *69*, 4208–4220.

[20] Binder, M. J.; Bashore, F. M.; Dunn Hoffman, K. K.; et al. Correction to "Cellular context influences kinase inhibitor selectivity". *J. Med. Chem.* **2026**, *69*, 20069–20071.

[21] Harris, C. R.; Millman, K. J.; van der Walt, S. J.; et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362.

[22] McKinney, W. Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference. 2010; pp 56–61.

[23] Virtanen, P.; Gommers, R.; Oliphant, T. E.; et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **2020**, *17*, 261–272.